

What's new on NoiseModelling 6.0?

Nicolas Fortin

NoiseModelling Days
September 24, 2026

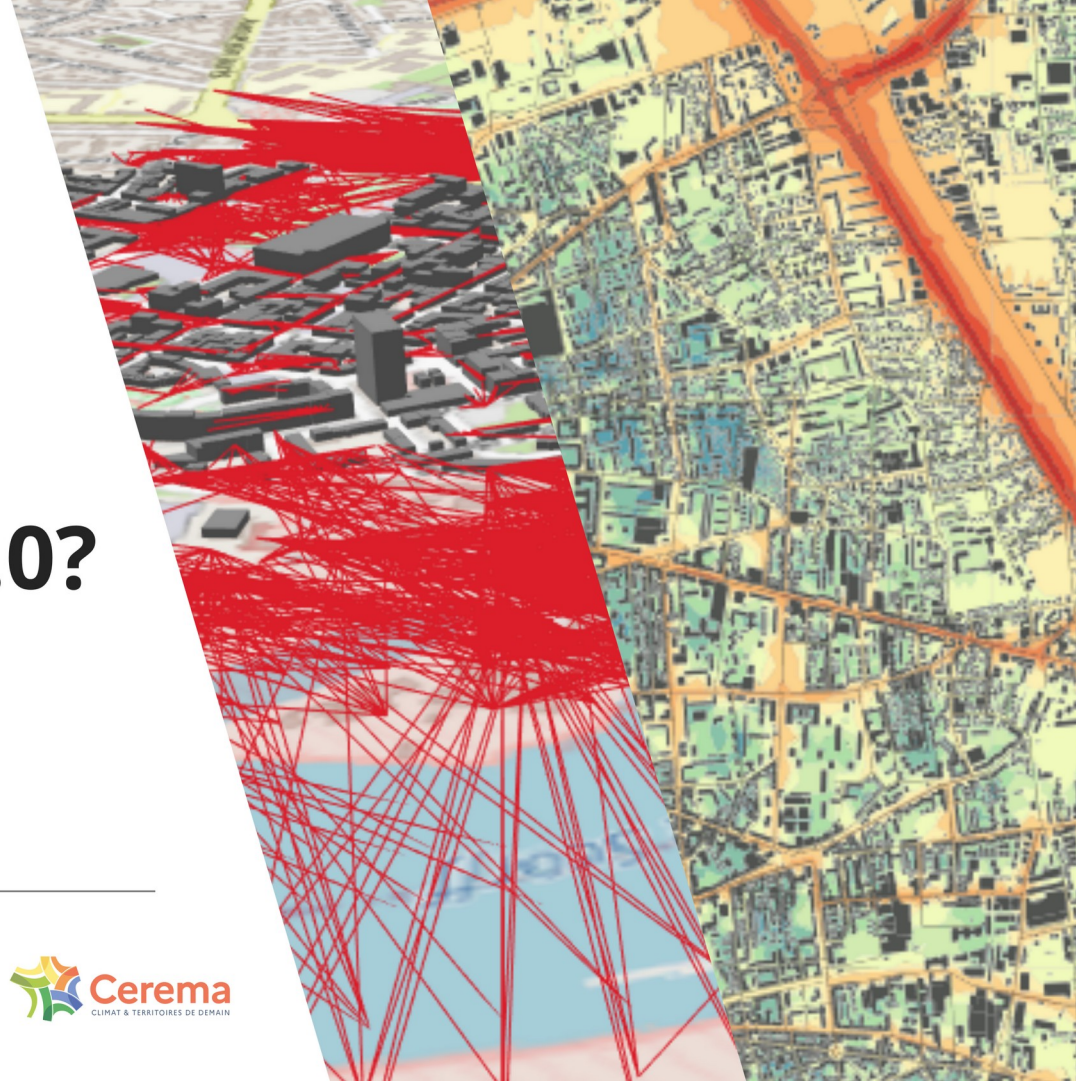
Organized by



Université
Gustave Eiffel



Cerema
CLIMAT & TERRITOIRES DE DEMAIN



Contents

A refreshed user interface

User side

- CNOSSOS ISO compliance
- New functionalities

Developer side

- Code refactoring
- Automated publishing

Server side

- New hosted mode multi-user NoiseModelling web GUI



New Graphic User Interface

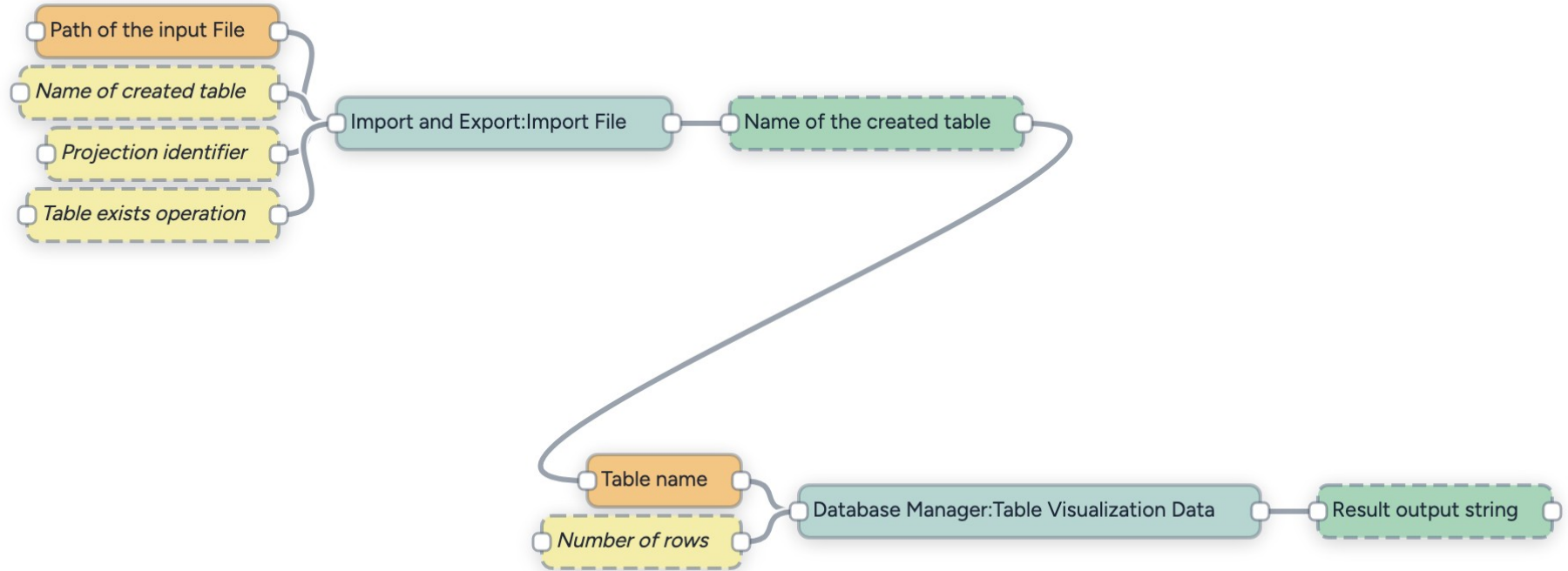
New template (a .css file that you can customize)

The screenshot displays the NoiseModelling GUI interface. At the top is a dark blue header bar with the 'NoiseModelling' logo on the left and buttons for 'File', 'Clear', 'Help', 'Manage', and 'Run Process' on the right. Below the header, the interface is divided into three main sections:

- Left Panel (Tools):** A sidebar with a search filter and a list of tools categorized under 'Acoustic Tools', 'CBS', and 'Data Assimilation'.
 - Acoustic Tools:** Add Laeq Leq columns, Create Isolines, Create Isosurface, DynamicIndicators.
 - CBS:** ComputePerDepartment, ComputePerUUeID, ExposeDEMforUniqueUUeID, Generate receivers, Generate sources, Write PostGIS Settings.
 - Data Assimilation:** All Possible Configuration, Create Assimilated Maps, Data Simulation, Extract Best Configuration, Merged Sensors Receivers, NMs 4 BestConfigs, Prepare Sensors.
- Center Panel (Builder):** A workflow editor with tabs for 'Builder' and 'Python'. It contains a visual flowchart of processes:
 - Top Flow:** 'Projection name' (input) connects to 'CBS:Generate sources' (process), which connects to 'Result output string' (output).
 - Middle Flow:** 'Display columns of the tables' (input) connects to 'Database Manager:Display Database' (process), which connects to 'Result output string' (output). Below this, 'SQL queries' and 'Output format' (inputs) connect to 'Database Manager:Execute Query' (process), which also connects to 'Result output string' (output).
 - Bottom Flow:** A group of inputs ('Configuration identifier', 'Projection name', 'UUeID pattern', 'HPC Configuration Name', 'Slurm task count', 'SSH Private Key password') connects to 'CBS:ComputePerUUeID' (process), which connects to 'Result output string' (output). To the right, 'Table name' and 'Number of rows' (inputs) connect to a process (partially obscured) which connects to 'Result output string' (output). Below this, 'Name of the table' and 'Projection identifier' (inputs) connect to 'Database Manager:Table Visualization Map' (process), which connects to 'Result output' (output). At the very bottom, another 'Name of the table' and 'Projection identifier' (inputs) connect to another 'Database Manager:Table Visualization Map' (process), which connects to 'Result output geom' (output). A 'Buildins table name' input is also visible at the bottom.
- Right Panel (Inputs/Console):** A section with tabs for 'Inputs' and 'Console'. Below the tabs is a map visualization showing a city street grid with a large red-shaded area indicating a specific geographic region.

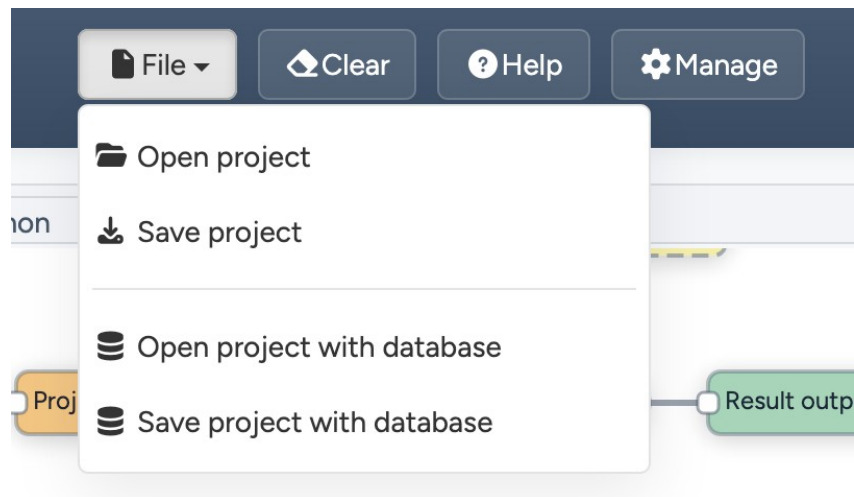
New Graphic User Interface

Chain processing by linking the blocks



New Graphic User Interface

- A search engine that works!
- When you move a block, all the inputs/outputs follow!
- Backup/Restore your project with the Save/Open menu items



New Graphic User Interface

New job status page

Manage jobs

Processing job management

Delete All

Job ID: 792

User	admin@localhost
Script	Database_Manager:Table_Visualization_Map
Start Date	2026-09-17 07:53:40
End Date	2026-09-17 07:53:41
Duration	0h 0m 0s
Progression	100 %
Status	COMPLETED
Logs	View Logs

Cancel Delete

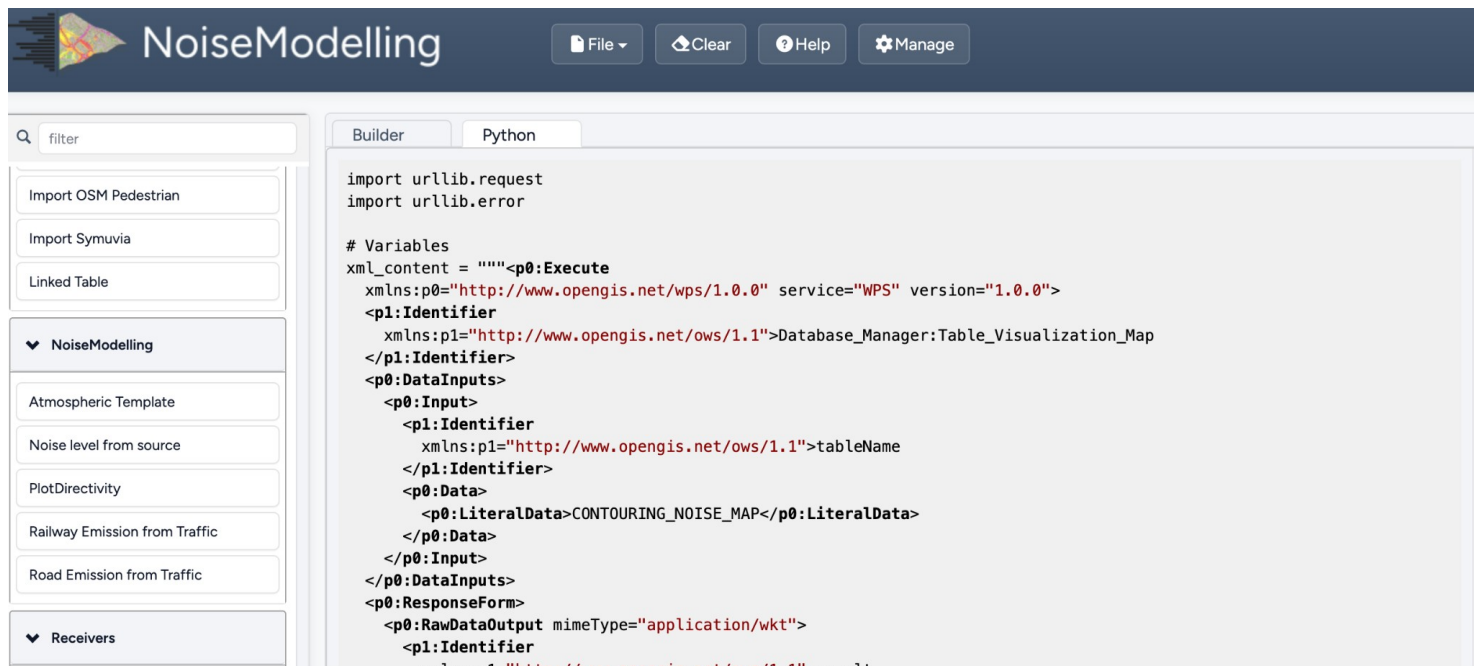
Job ID: 791

User	admin@localhost
Script	Database_Manager:Display_Database
Start Date	2026-09-17 07:53:33
End Date	2026-09-17 07:53:33
Duration	0h 0m 0s

New Graphic User Interface

Python script tab

Once a block has been executed, a Python script is automatically generated to help you automate your processes.

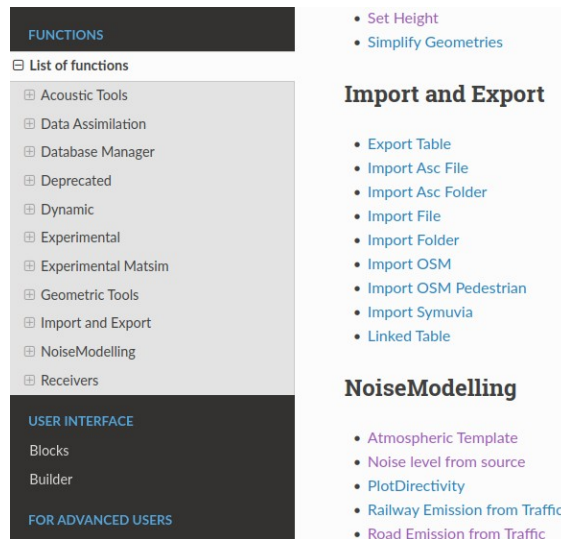


User side news

Documentation updated, with all functions added →

New blocks:

- **Data assimilation** : combines observations with a numerical model to improve the accuracy of forecasts or analyses. New tutorial
- **Linked Table** : Import data directly from an external database like PostGIS
- **IsoLines** : Alternative to iso surfaces, contribution from Ignacio Soto Molina, thanks !
- **Execute Query**: Directly run SQL queries in your database



User side news

NoiseModelling CLI

Experimental, run blocks directly through a PostGIS connection instead of a local H2GIS database.

NoiseModelling API

Compatibility with the Python library *owslib.wps*

Dev side news

Code refactoring:

- Reorganization of package / class
- Optimization & Fix with more unit tests

Automated publishing:

- Packaging for Windows / MacOS / Linux
- Nightly Docker / PodMan container image upload

Server mode

NoiseModelling several deployment options:

When run locally:

- Single default user without authentication

Hosted on a server:

- Multiple users with login / One Time Password (TOTP)
- One database per user
- Administrator or basic user profile
- Easy & Secure deployment thanks to docker / podman / duplicity

Thanks for your attention

Any questions ?

